# Latent Variable Models
# Estimation

# Topics

- Non-normal data
- Ordinal Data
  - Polychoric Correlations
  - DWLS
- Small sample adjustments
- Missing Data

# Factors Affecting Covariance

Covariance is the “data” for SEM, and it is easy to overlook the impact that individual observations can have on the results because individual observations are not included in the “data”

Two important factors can impact the validity of the use of covariance:

- Outliers
- Nonlinearity

# Outliers

Any observation more than 3 SD from the mean on any variable

**Impact**

- Have a large effect on the mean
- Contribute a large deviation

**Reasons/Solutions**

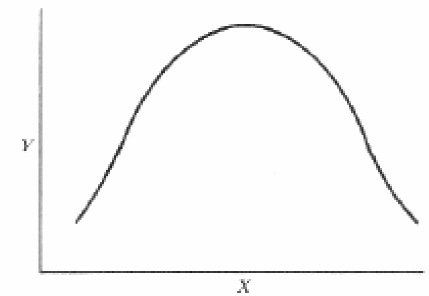
- Measurement Error
  - Can sometimes detect and fix
- Multiple Populations
  - Exclude deviant (small) groups in recruitment
  - Oversample deviant groups
- Random Values
  - With large samples come large deviations
  - Trimming or Windsorizing

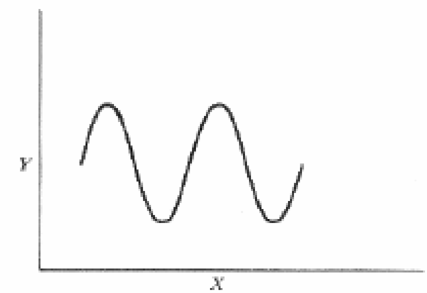
# Non-Linearity

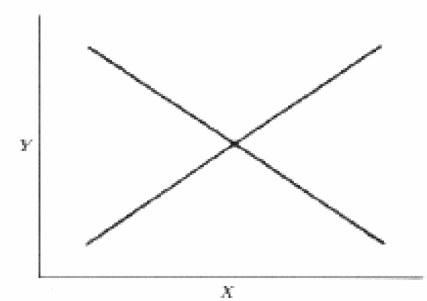
Covariance is a measure of **linear** association

Different types of non-linearity may require different solutions

- Curvilinear
- Cyclical
- Interactions



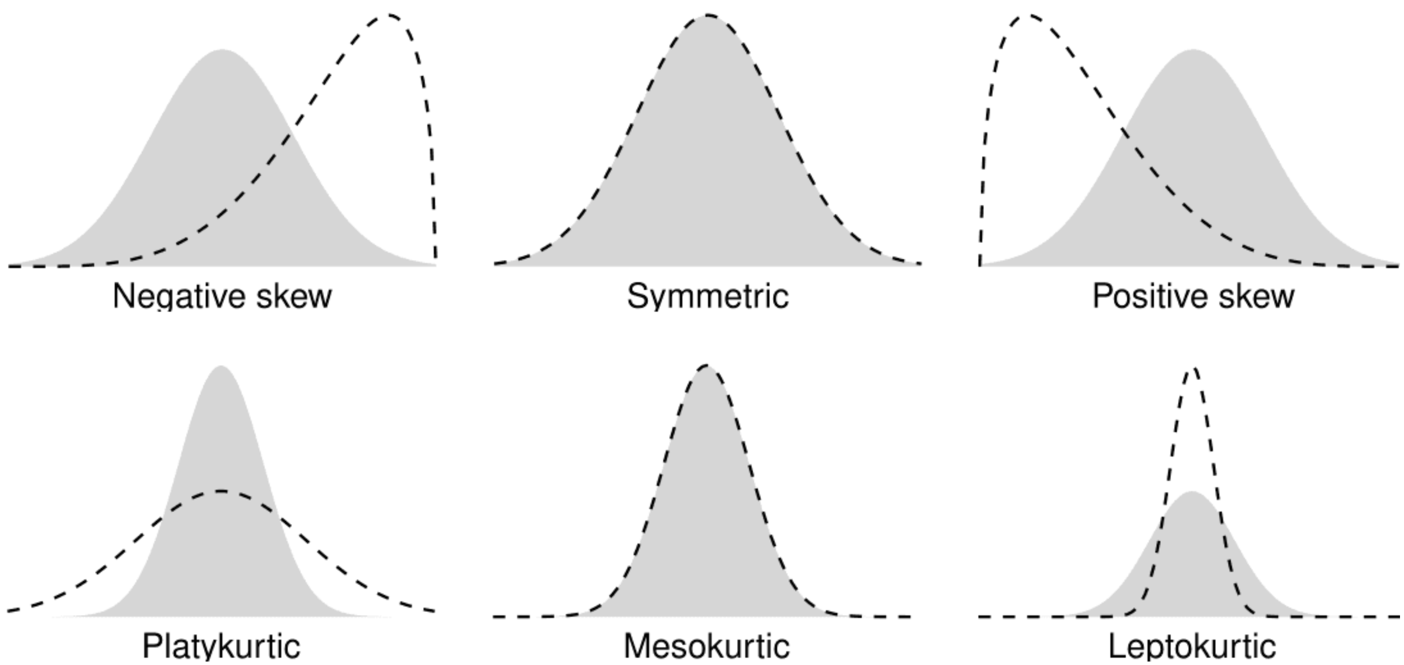




Fife, D. A., Brunwasser, S. M., & Merkle, E. C. (2022, January 27). Seeing the Impossible: Visualizing Latent Variable Models With Flexplavaan. *Psychological Methods*. http://dx.doi.org/10.1037/met0000468

# Importance of Assumed Normality

- For multivariate normal distributions, mean vector and covariance matrix from the data are sufficient statistics
  - These represent the first and second moments of the multivariate distribution
- Skew and kurtosis are zero for normal distribution

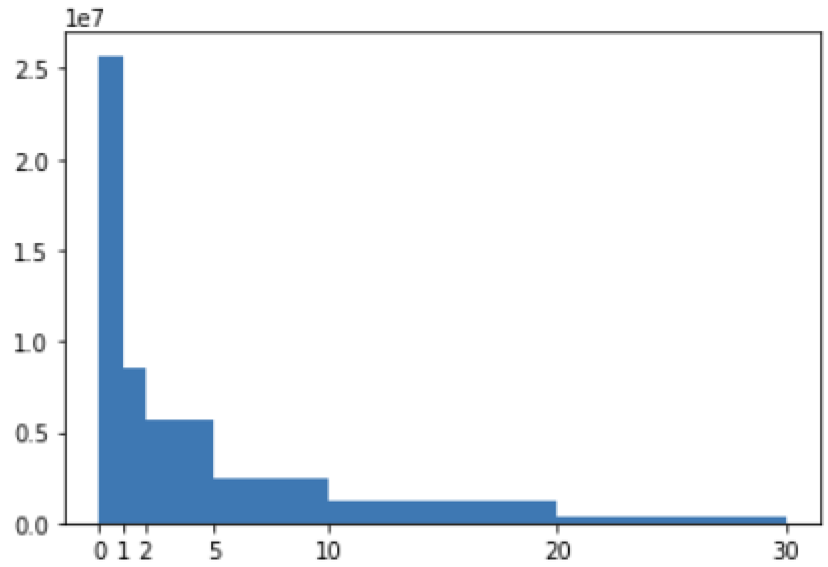


# Maximum Likelihood

ML estimation explicitly relies on the multivariate normal probability density function when constructing the likelihood

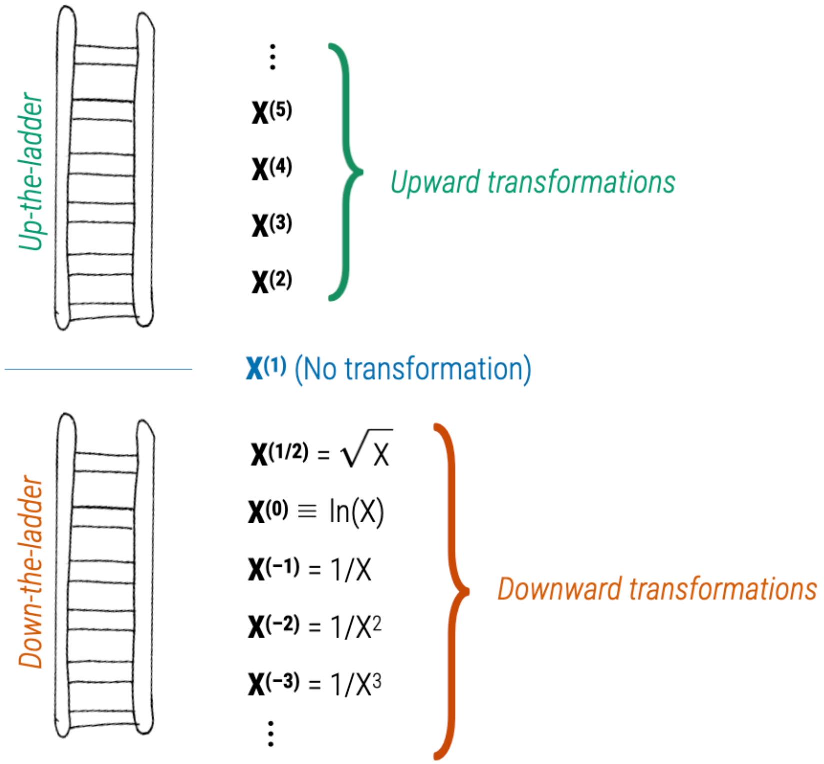
$$f(\boldsymbol{z}|\boldsymbol{\mu},\boldsymbol{\Sigma}) = (2\pi)^{-(N_Z)/2}\det(\boldsymbol{\Sigma})^{-\frac{1}{2}}\,\mathrm{e}^{-\frac{1}{2}(\mathbf{z}-\boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\mathbf{z}-\boldsymbol{\mu})}$$

So what is one to do if the observed data is not normally distributed?



# Options: Transformations

Sometimes power transformations can be used to meet assumptions of normality




But these transformations can often fail, especially when you have zero inflated data (many zero observations)

# Options: Ignore Non-Normality

Non-normality is only an issue for endogenous variable

Exogenous variables can be binary, ordinal, count, etc., without issue

Maximum likelihood parameter estimates are still consistent (i.e., converges to the population value as sample size increases)

- The only exception to this is if kurtosis gets too large

ML standard error estimates will be biased if there are violations of normality in the endogenous variables

- ML estimates are not asymptotically efficient
- Typically SEs are too small
  - CIs are too small
  - Type I Error is too large
- Typically $\chi^2$ is too large
  - Models are over-rejected

# Robust Standard Errors

Standard errors can be corrected for non-normality based on characteristics of the sample data

These are often called “robust SEs”, “Satorra-Bentler”, or “sandwich estimator” SEs

Still using Maximum Likelihood Estimates, but adjusting the SEs

Can apply a similar adjustment to the likelihood ratio test statistic

These approaches can be easily used in lavaan with the following options

- `se = “robust”`
- `test = “robust”`

# Bootstrapping

An alternative non-parametric approach is bootstrapping

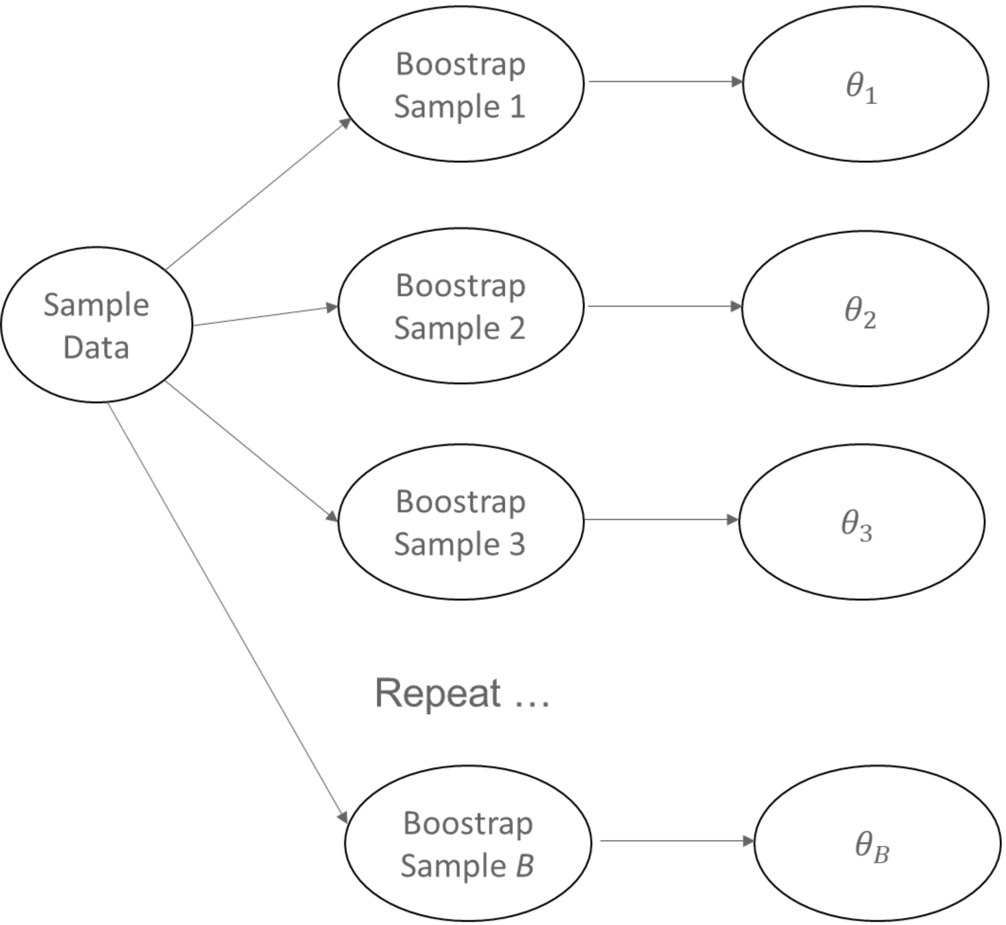
- Sample N cases from original sample with replacement.
- Estimate the model in “resample”
- Repeat over and over
- Calculate the standard deviation of resampled estimates
- This is used as the bootstrap SE

Given a bootstrap SE, can calculate a bootstrap p-value etc.

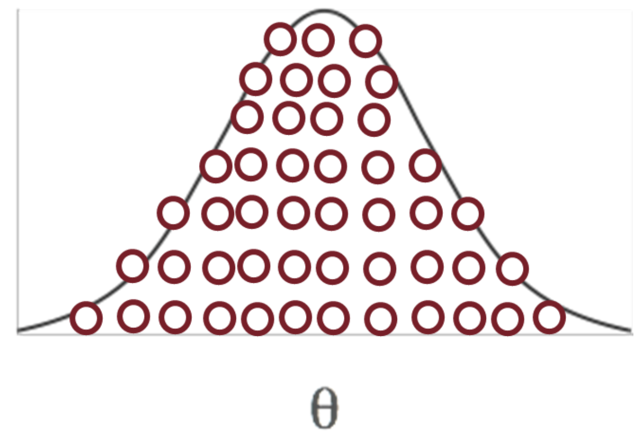
This approach can be easily used in lavaan with the following options

- `se = “bootstrap”`
- `test = “bootstrap”`

# Visualization of Bootstrap




Standard error of this distribution = boostrap SE.




$$SE_{Boostrap} = \sqrt{\frac{\sum_{b=1}^{B}\left(\hat{\theta}_b - \bar{\theta}\right)}{B-1}}$$

Cf. Enders, 2015

# Latent Factors with Categorical Indicators

- Starting in our very first class, we learned that SEM is just a bunch of regressions.

- Categorical (e.g., binary, ordinal, count) are easily entered as DV's (endogenous vars) in path analysis (just declare them as CATEGORICAL in Mplus or ordered in lavaan).

- But what if categorical items are used as indicators in a factor analysis?

- SEM methods have been developed for CFA and structural regression with categorical indicators.

- Today we will review CFA methods for binary and ordinal indicators.

# Issues with Categorical Indicators

- In **model world**:
  - Factor loadings may be attenuated.
  - ↓SEs, which risk ↑Type I errors.
  - Further, you may end up with inadmissible predicted (e.g., y-hat) values, such as scores < 0 or >1 on a 0/1 variable.
  - Fit statistics may also be affected. (See Finney & DiStefano, 2013).

- In **sample/covariance world**, the covariance matrix of your binary/ordinal variables (if treated continuous) may underestimate the true variability in the underlying construct being measured.

- Thus, binary and ordinal indicators present challenges both in terms of:
  - How to incorporate them into the SEM model, and
  - How best to model the covariation among discrete variables.

# Issues with Categorical Indicators

- The problems with treating ordinal variables as continuous increase when there are fewer response categories and/or when the variables are non-normally distributed.

  - If ≥ 5 ordered categories *and* approx. normal distributions → can treat as continuous with limited problems.

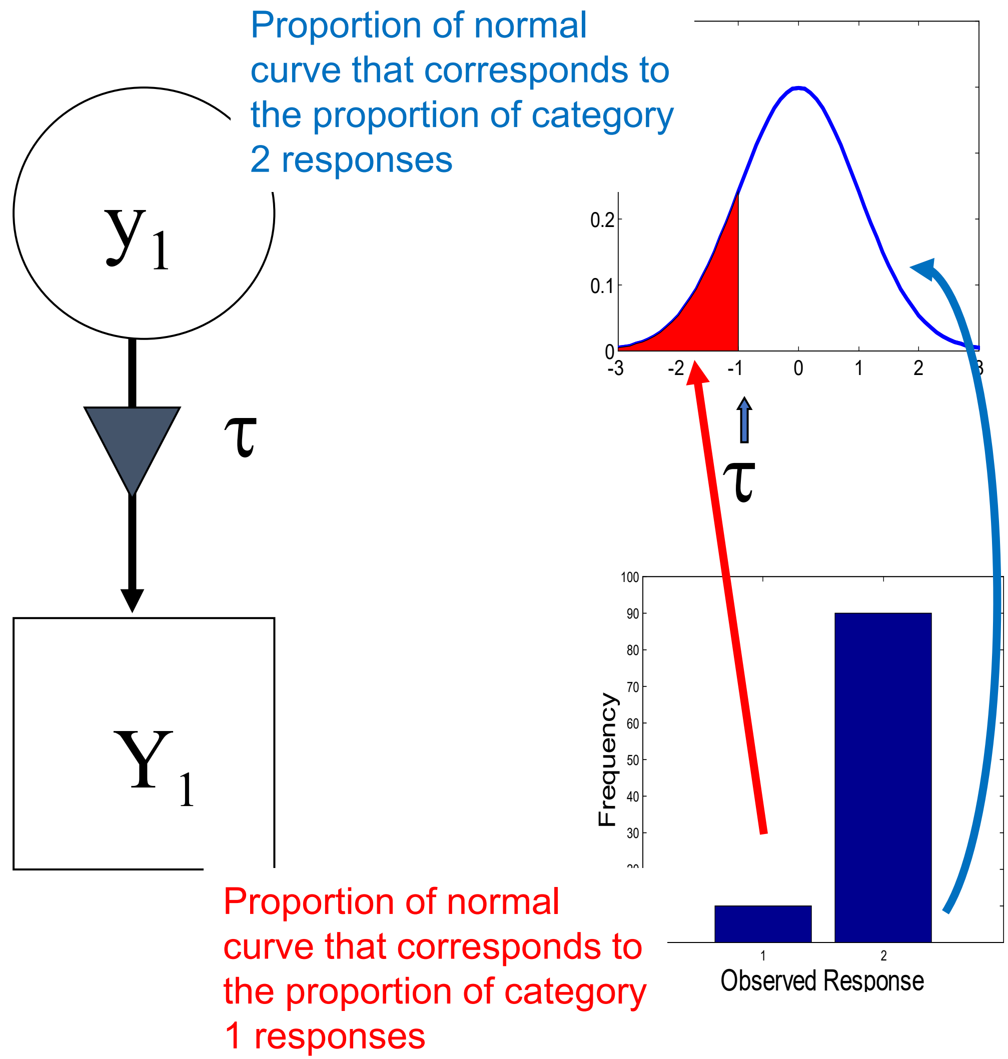
  - If < 4 or 5 ordered categories *or* with nonnormal distributions → alternative methods are needed.

# Discrete (binary) model of underlying response

**Normal Score:**
**Z score,**
***m* = 0, *s* = 1**
**by assumption**
**and used to**
**assure**
**identification**

**This path symbol is a "rectifier"**

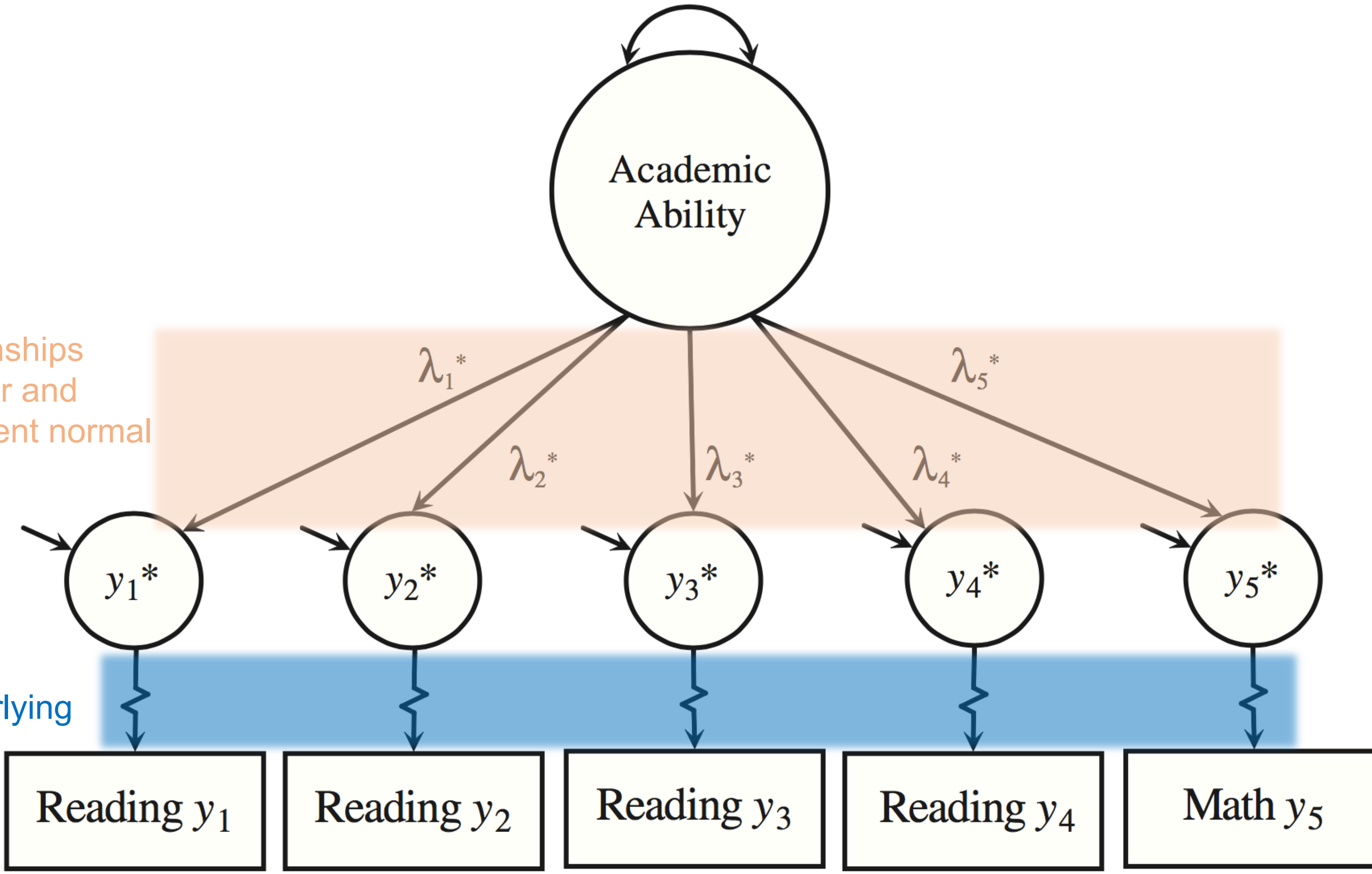
**Observed Binary Response:**
**Y=0 or Y=1**



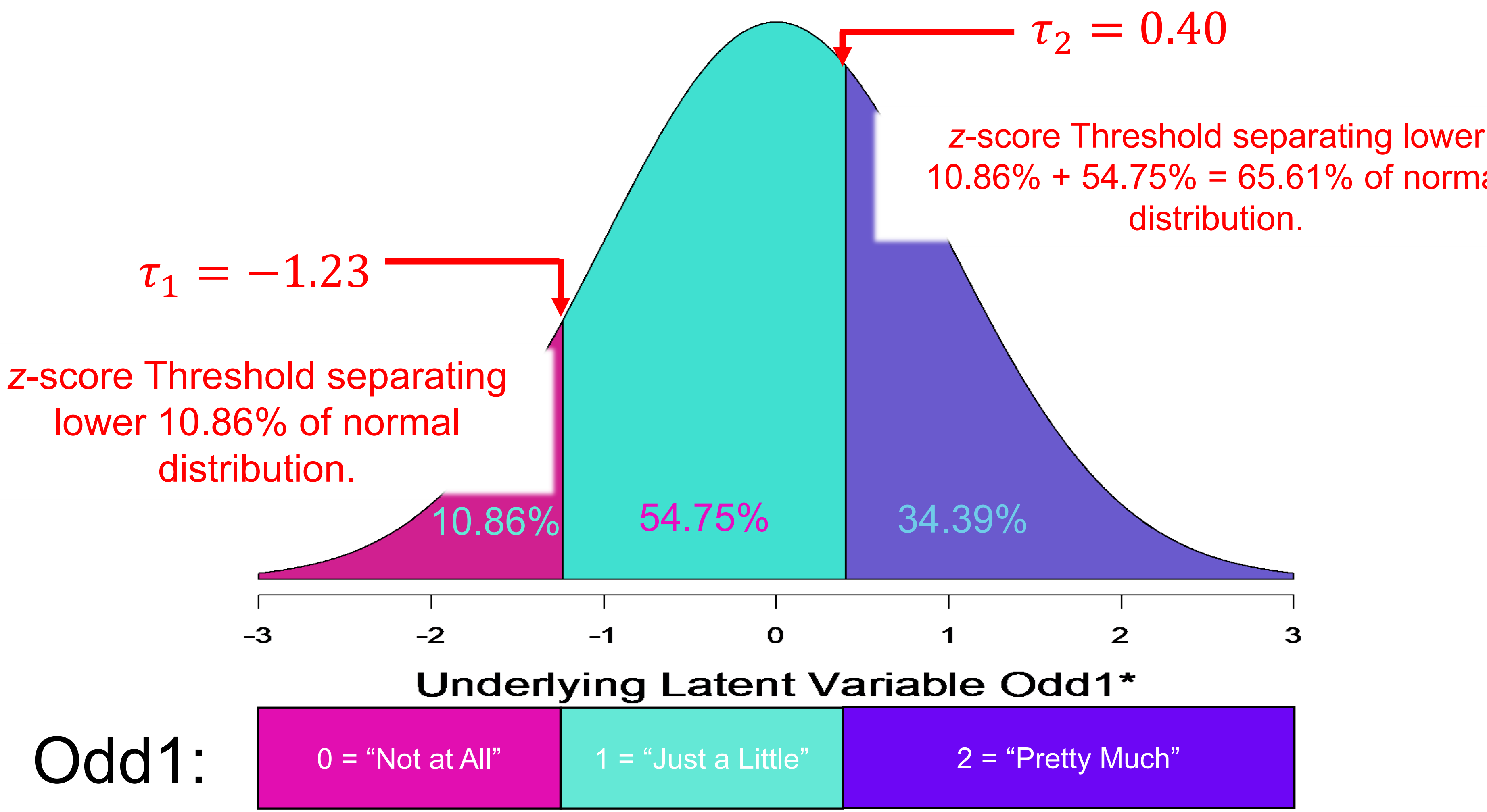
# Edwards, Wirth, Houts, & Xi, 2012

Linear relationships between factor and underlying latent normal variables.

Nonlinear relationships between underlying latent normal variables and categorical indicators.



# Generalizes to More Categories



# Estimating Sample Thresholds

- To estimate thresholds in the sample, we first calculate the proportion of the total sample of participants who have selected a particular category *c* on the categorical variable.

$$p_c = \frac{N_c}{N}$$

- To calculate the appropriate threshold, we utilize the **inverse normal distribution function**, $\Phi^{-1}(\text{probability below } z, \mu = 0, \sigma = 1)$.

- This function takes as **input**:
  - A probability (proportion).
  - A mean (default assumed = 0 for standard normal).
  - A standard deviation (default assumed = 1 for standard normal).

- And **returns** a *z*-score (quantile). The area of the normal curve *below* this *z*-score (threshold) equals the probability entered into the function.

- In R this is accomplished via **`qnorm(p, mean = 0, sd = 1, lower.tail = T)`**.

# Estimating Sample Thresholds

- Note: this function is the inverse of the **cumulative normal distribution function**, $\Phi(z, \mu = 0, \sigma = 1)$, which takes as **input**:

  - A *z*-score (quantile).
  - A mean (default assumed = 0 for standard normal).
  - A standard deviation (default assumed = 1 for standard normal).

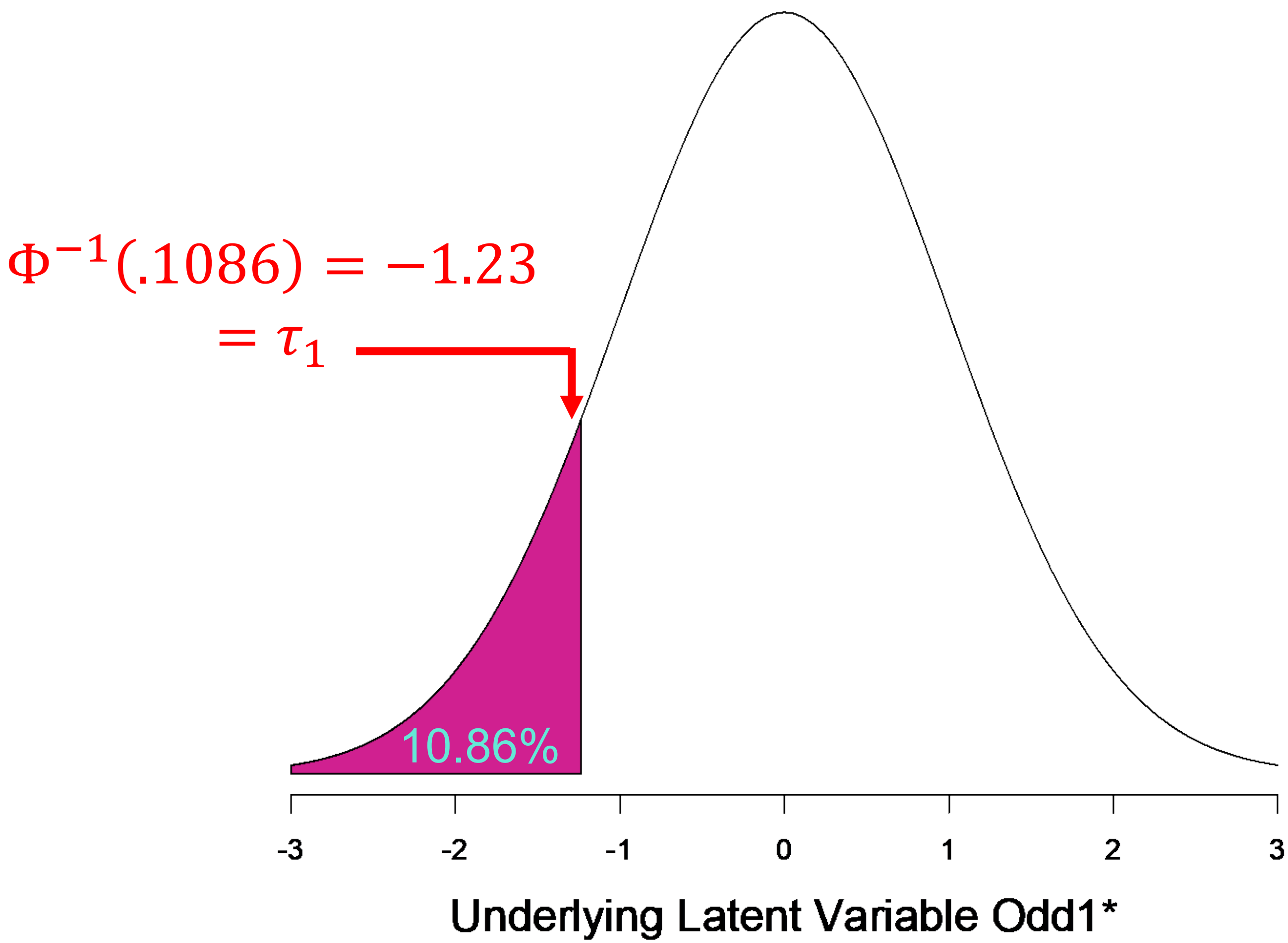
- And **returns** a probability (proportion). This proportion corresponds to the area of the normal curve *below* the *z*-score (threshold) entered into the function.

- In R this is accomplished via **`pnorm(q, mean = 0, sd = 1, lower.tail = T)`**.

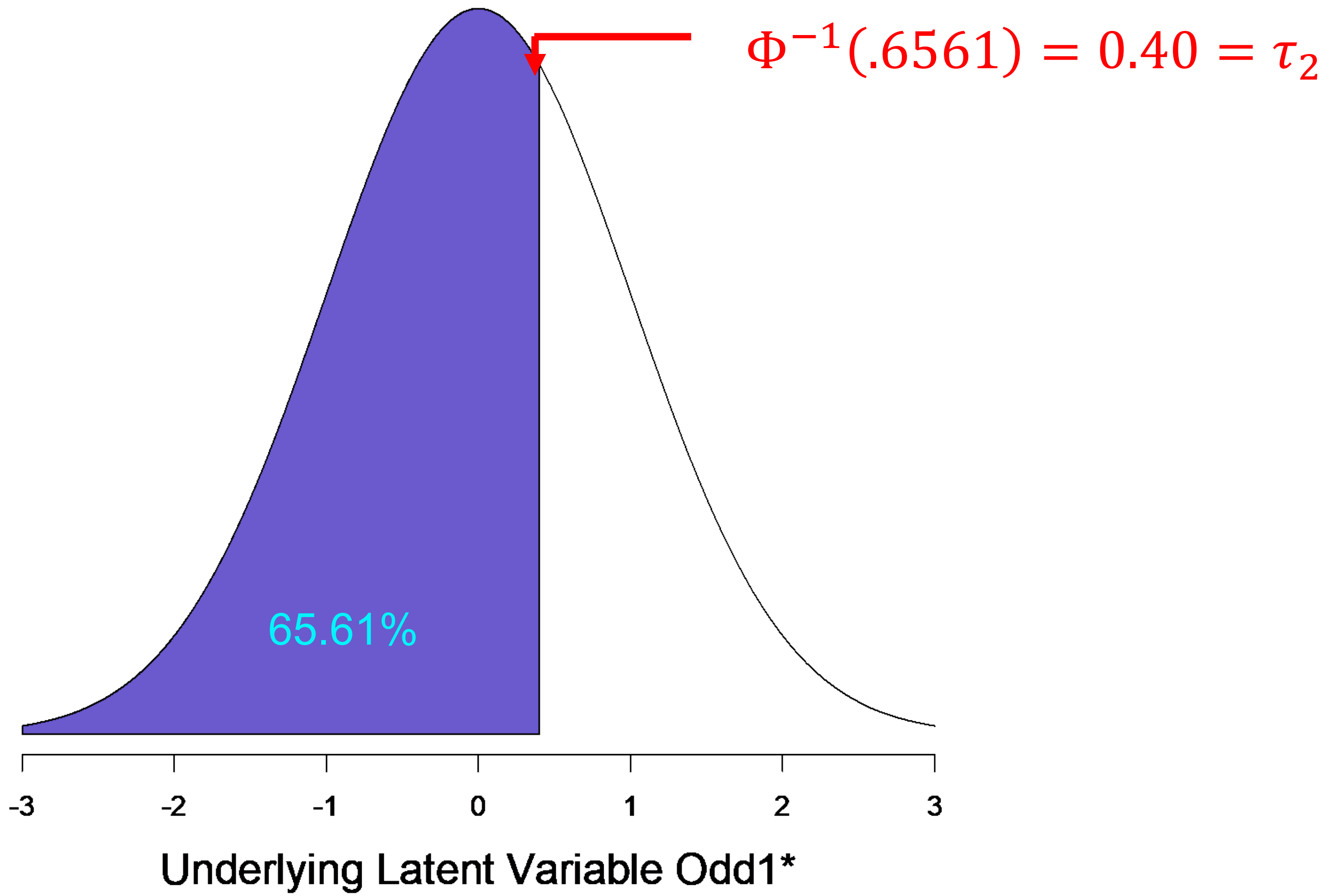
- With this in mind, we can use the inverse distribution function $\Phi^{-1}$ (`qnorm()`) to estimate thresholds given observed response proportions.

- We can also calculate the proportion of the normal curve corresponding to a given (proposed) threshold using the cumulative distribution function $\Phi$ (`pnorm()`).

# Underlying Latent Normal Distribution



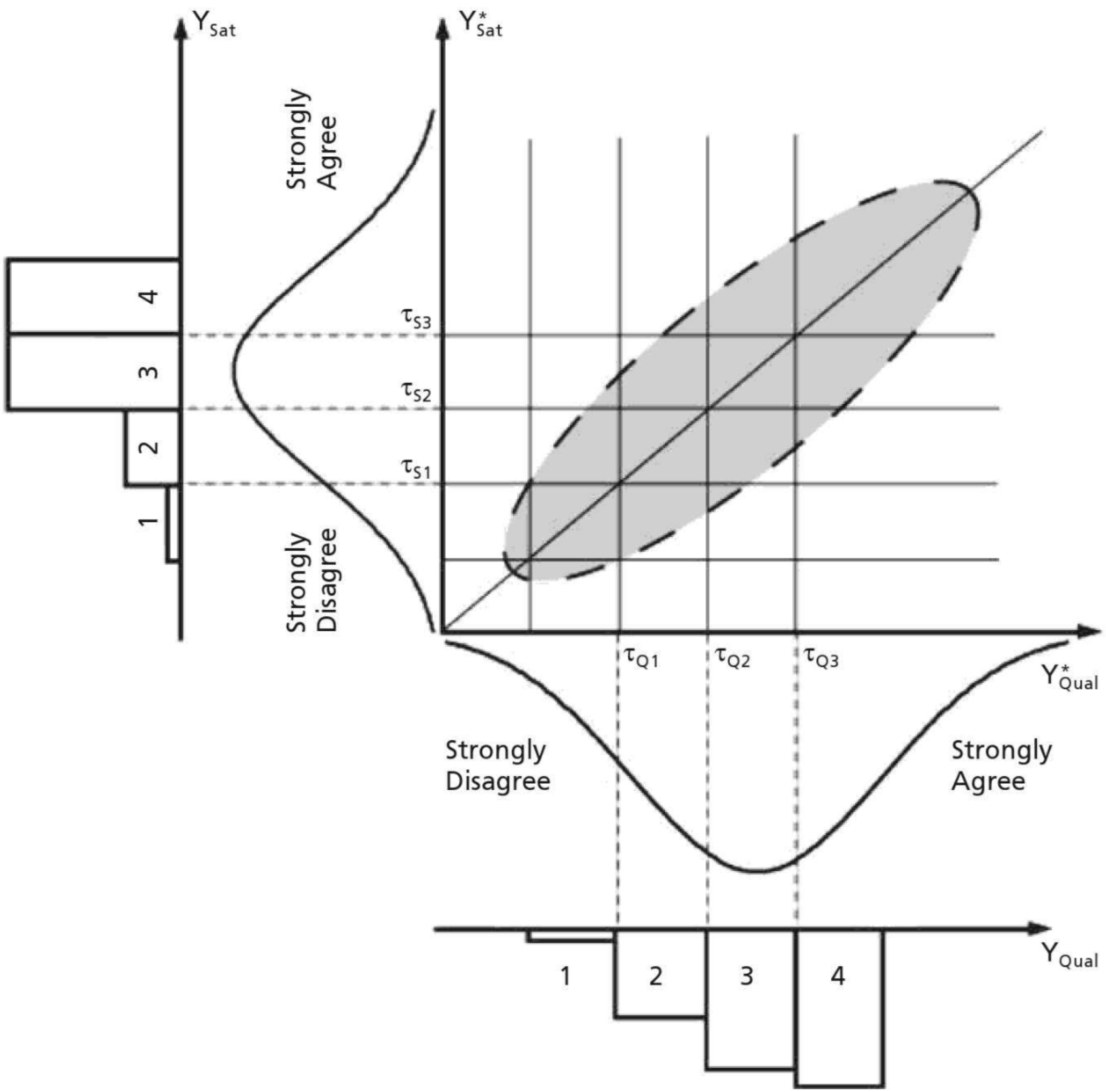
# Underlying Latent Normal Distribution



# Using Thresholds to Estimate Categorical Variable Correlations

- Based on the assumption of underlying normals …
- **Tetrachoric Correlations** estimate the correlation between two dichotomous variables.
- **Polychoric Correlations** estimate the correlation between two ordinal variables.
- **Polyserial Correlations** estimate the correlation between a categorical (binary/ordinal) and continuous variable.

# Illustration: Polychoric Correlation




Finney & DiStefano, 2013

# Why Is This Relevant?

- Recall that all of SEM involves comparing model-estimated parameters (covariances, means) to observed sample statistics (sample covariances, means).

- With polychoric and tetrachoric correlations in hand, we have **viable sample statistics for our categorical variables** that can be used as a basis for comparison against our model estimates!

Polychoric Correlations (Above Diagonal)

Pearson's *r* (Below diagonal)

| | odd1 | odd2 | odd3 | odd4 | odd5 | odd6 | odd7 | odd8 |
|---|---|---|---|---|---|---|---|---|
| odd1 | 1.00 | 0.64 | 0.54 | 0.34 | 0.22 | 0.39 | 0.48 | 0.42 |
| odd2 | 0.51 | 1.00 | 0.53 | 0.28 | 0.18 | 0.34 | 0.42 | 0.38 |
| odd3 | 0.42 | 0.41 | 1.00 | 0.41 | 0.12 | 0.25 | 0.51 | 0.44 |
| odd4 | 0.26 | 0.21 | 0.31 | 1.00 | 0.38 | 0.31 | 0.3 | 0.55 |
| odd5 | 0.17 | 0.14 | 0.09 | 0.29 | 1.00 | 0.45 | 0.3 | 0.33 |
| odd6 | 0.29 | 0.27 | 0.19 | 0.24 | 0.35 | 1.00 | 0.49 | 0.42 |
| odd7 | 0.32 | 0.3 | 0.34 | 0.22 | 0.22 | 0.36 | 1.00 | 0.56 |
| odd8 | 0.28 | 0.26 | 0.31 | 0.4 | 0.23 | 0.29 | 0.41 | 1.00 |

# Why Is This Relevant?

- I.e., instead of:

$$\boldsymbol{\Sigma}(\boldsymbol{\theta}) - \mathbf{S}$$

- With a model of entirely ordinal variables, we have:

$$\mathbf{R}^{*}_{\text{Model Implied}}(\boldsymbol{\theta}) - \mathbf{R}_{\text{Polychoric}}$$

- The actual implementation of this is a slightly fancier weighted least squares (WLS) function, but this is the general conceptual idea.

- Note that we are using polychoric/tetrachoric **correlations** instead of **covariances** here, simply because these underlying normal variables have no explicit scale. So they don't have known variances to model.

# Estimation

- Categorical Variable SEMs are typically estimated using some variation on Weighted Least Squares (WLS) estimation:

$$F_{WLS} = \left(\boldsymbol{s}_{ij} - \hat{\boldsymbol{\sigma}}_{ij}\right)' \mathbf{W} \left(\boldsymbol{s}_{ij} - \hat{\boldsymbol{\sigma}}_{ij}\right)$$

- Where **s** is a vector of sample thresholds and categorical (e.g., polychoric) correlations, $\hat{\boldsymbol{\sigma}}$ is the model implied version, and **W** is a weight matrix used in estimation (ACOV of **s**, fwiw…).

This implies that the fit has to do with the squared residuals (discrepancy) between the observed and implied correlations (and thresholds!)

The **W** matrix can be large and problematic, so programs often use only the diagonal elements – **D**iagonally **W**eighted **L**east **S**quares (DWLS).

# Implementation in Lavaan

The most direct way to specify data as ordinal with the ordered option:

```
fit <- cfa(myModel, data = myData, ordered = c("item1","item2",
"item3","item4"))
```

If all the (endogenous) variables are to be treated as categorical, you can use ordered = TRUE as a shortcut.

When the ordered= argument is used, lavaan will automatically switch to the WLSMV estimator: it will use diagonally weighted least squares (DWLS) to estimate the model parameters, but it will use the full weight matrix to compute robust standard errors, and a mean- and variance-adjusted test statistic. Other options are unweighted least squares (ULSMV), or pairwise maximum likelihood (PML). Full information maximum likelihood is currently not supported.